

Digital Logic Design Final Project Report

Project Title: Traffic Light Controller with Pedestrian Request and Countdown Timer

Course: EECS 3201 F

Design Description:

This project implements a Traffic Light Controller with a pedestrian request system on the DE10-Lite FPGA using Verilog. Traffic lights are mapped to LEDR0–2 (Green/Yellow/Red), cycling through 5s Green, 2s Yellow, and 5s Red, with HEX0 showing the countdown. KEY1 can be pressed at any time; the request is stored and only activates during the next Red phase. Pressing during Green or Yellow extends the upcoming Red to 9s, while pressing during Red defers the crossing to the following Red. During the pedestrian phase, LEDR3 blinks at 1 Hz and HEX0 displays a 9-second countdown. After completion, the request clears and the controller returns to the normal $G \rightarrow Y \rightarrow R$ cycle.

The project's testable features include:

- Automatic Green/Yellow/Red sequencing
- Accurate 5s, 2s, and 5s timing
- Real-time countdown on HEX0 for each phase
- Safe, synchronized detection of KEY1 pedestrian requests
- Queuing behavior that mimics real traffic intersections
- Extended 9-second Red time when pedestrians are active
- Blinking LEDR3 pedestrian walk signal
- Full system reset functionality

Sequential logic implements the FSM, counters, button edge detection, and blink generator. Combinational logic drives LED outputs, pedestrian behavior, and seven-segment decoding. The system operates like a realistic traffic intersection with a safe, predictable pedestrian crossing mechanism.

Reference:

I drew inspiration from a YouTube tutorial on traffic light controllers, which helped me better understand how to structure the FSM and timing logic before adapting the concepts to our own design.

- YouTube — “*Counters (Part 7) - Traffic Light Controller FSM (.vhd)*”, S8UvMZuBy8c. Available at: <https://www.youtube.com/watch?v=S8UvMZuBy8c>